

## Highlights

### **The Beggining of Control Revolution: Offset-free Koopman MPC**

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- Research highlight 1
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# The Beggining of Control Revolution: Ofset-free Koopman MPC

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## Abstract

*Keywords:*

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## 1. Introduction

## 2. Preliminaries and Notation

### 2.1. Offset-Free MPC

In this subsection, we will present a standard offset free model predictive control framework, introduced by [TODO: cite pannocchia](#) . This approach consist of state estimation - observer, target optimization and dynamic optimization. The whole approach is used to introduce integral action into model predictive control to help mitigate plant-model mismatch and ensure offset free control. If this compensation would not be used, the plant with this controller would not reach a desired reference (other than at origin), due to predictions using different model than actual plant.

The offset-free augmented system can be written in the following state-space form, including the disturbance input matrix  $B_d$  and disturbance output matrix  $C_d$ :

$$\begin{bmatrix} x_{k+1} \\ d_{k+1} \end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} x_k \\ d_k \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u_k, \quad y_k = \begin{bmatrix} C & C_d \end{bmatrix} \begin{bmatrix} x_k \\ d_k \end{bmatrix} \quad (1)$$

where  $x_k$  is the original system state,  $d_k$  is the estimated constant disturbance,  $u_k$  is the control input,  $y_k$  is the output,  $A$ ,  $B$ , and  $C$  are the original state-space matrices,  $B_d$  and  $C_d$  encode how the disturbance enters the state and output equations, and  $I$  is the identity matrix. Here,  $n_x$ ,  $n_u$ ,  $n_y$ , and  $n_d$  denote the state, input, output, and disturbance dimensions, respectively, so

that  $x_k \in \mathbb{R}^{n_x}$ ,  $u_k \in \mathbb{R}^{n_u}$ ,  $y_k \in \mathbb{R}^{n_y}$ ,  $d_k \in \mathbb{R}^{n_d}$ ,  $A \in \mathbb{R}^{n_x \times n_x}$ ,  $B \in \mathbb{R}^{n_x \times n_u}$ ,  $C \in \mathbb{R}^{n_y \times n_x}$ ,  $B_d \in \mathbb{R}^{n_x \times n_d}$ ,  $C_d \in \mathbb{R}^{n_y \times n_d}$ , and  $I \in \mathbb{R}^{n_d \times n_d}$ . For the offset free behavior, the number of the disturbances  $n_d$  must be lower or equal to the number of the measurements  $n_y$ . Also it is necessary to note that the number of offset free tracking is achievable only in the number of disturbances that are estimated. For the system where  $n_d = n_y$  system and choice of the matrices  $B_d$  and  $C_d$  there is a necessary condition, that:

$$\text{rank} \begin{bmatrix} I - A & -B_d \\ C & C_d \end{bmatrix} = n_x + n_y. \quad (2)$$

When solving the MPC, both states and disturbance are approximated by observer.

## 2.2. Koopman Operator

The koopman operator theory was introduced by [TODO: cite](#) and further popularize by [TODO: cite](#). The theory enables to exactly model nonlinear systems linearly, by lifting its current states/measurements into infinite dimensional variables, where the governing operator for the evolving system is linear.

[TODO: fancy smenši exuation](#)

Over the years many approximations of the koopman operator from the data were proposed, starting with simple DMD, extended to the EDMD and recently the neural networks were used to best approximate the lifting functions so the system behaves linearly. The algorithms were as well developed not only for autonomous systems but for the systems with control as well [TODO: cite and elaborate](#). Koopman operator in the MPC was introduced by [TODO: cite Korda](#), and in recent years almost every kind of MPC was transfer to accept some kind of Koopman Operator, mostly represented as an LTI system in the state space form. The offset free version is no exception, in the work of [TODO: Zotero, kukni papere, prosím ešte aj naskenuj ďalšie](#).

For the purpose of this work we are using the DeepKoopman Approximation introduced by [TODO: cite, Lusch](#), transformed for the nonautonomous system. We use two kinds of approximations. Firstly the LTI approximation and secondly the approximation, where only the lifted system evolves linearly, but its projection is nonlinear (as it should be from the definition of Koopman operator). The architectures of those systems can be seen below. [TODO: images](#)

### 3. Koopman MPC with Offset-Free

In this subsection, we will present a standard offset free model predictive control framework using approximation of Koopman operator as a linear time invariant (LTI) model. As observer is standard Kalman filter, we will discuss mainly remaining two optimization problems. As we use the LTI approximation of Koopman operator, we can use the standard MPC formulation with linear constraints and objective function. The control structure is shown in Figure 1.

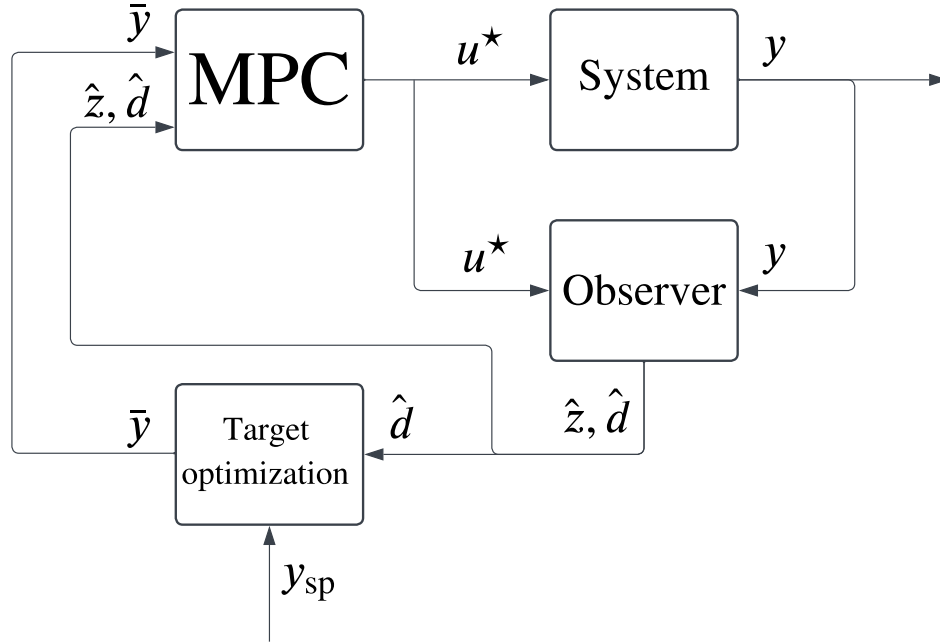


Figure 1: Closed-loop control structure. **TODO:** prerobiť tak aby znormalizoval fonty a haty, rozdeliť d a z

The target optimization problem computes the steady-state targets  $\bar{u}$ ,  $\bar{z}$ ,  $\bar{y}$  that minimizes the difference between reference  $y_{\text{ref}}$  and feasible output  $\bar{y}$  in steady state. The problem is formulated as:

$$\min_{\bar{u}, \bar{z}, \bar{y}} (\bar{y} - y_{\text{ref}})^\top Q_y (\bar{y} - y_{\text{ref}}), \quad (3a)$$

$$\text{s.t.} \quad \bar{z} = A\bar{z} + B\bar{u}, \quad (3b)$$

$$\bar{y} = C\bar{z} + d, \quad (3c)$$

$$u_{\min} \leq \bar{u} \leq u_{\max}, \quad (3d)$$

$$y_{\min} \leq \bar{y} \leq y_{\max}, \quad (3e)$$

$$d = \hat{d}(t), \quad y_{\text{ref}} = y_{\text{sp}}(t), \quad (3f)$$

where  $\bar{u} \in \mathbb{R}^{n_u}$ ,  $\bar{z} \in \mathbb{R}^{n_z}$ , and  $\bar{y} \in \mathbb{R}^{n_y}$  are the steady-state input, lifted state, and output, respectively;  $Q_y \succ 0$  is the output tracking weight matrix;  $A \in \mathbb{R}^{n_z \times n_z}$ ,  $B \in \mathbb{R}^{n_z \times n_u}$ ,  $C \in \mathbb{R}^{n_y \times n_z}$  are the state-space matrices of the LTI Koopman model;  $d = \hat{d}(t) \in \mathbb{R}^{n_y}$  is the additive output disturbance estimate provided by the Kalman filter;  $y_{\text{ref}} = y_{\text{sp}}(t)$  is the output setpoint; and  $u_{\min}, u_{\max}, y_{\min}, y_{\max}$  are the input and output constraint bounds.

Given the steady-state target  $(\bar{u}, \bar{z}, \bar{y})$ , the MPC problem is solved at each sampling instant to compute the optimal input sequence over the horizon  $N$ :

$$\min_{u_0, \dots, u_{N-1}} \sum_{k=0}^{N-1} (y_k - \bar{y})^\top Q_y (y_k - \bar{y}) + \sum_{k=0}^{N-1} \Delta u_k^\top Q_u \Delta u_k, \quad (4a)$$

$$\text{s.t.} \quad z_{k+1} = Az_k + Bu_k, \quad k \in \mathbb{N}_0^{N-1}, \quad (4b)$$

$$y_k = Cz_k + d, \quad k \in \mathbb{N}_0^{N-1}, \quad (4c)$$

$$\Delta u_k = u_k - u_{k-1}, \quad k \in \mathbb{N}_0^{N-1}, \quad (4d)$$

$$u_{\min} \leq u_k \leq u_{\max}, \quad k \in \mathbb{N}_0^{N-1}, \quad (4e)$$

$$y_{\min} \leq y_k \leq y_{\max}, \quad k \in \mathbb{N}_0^{N-1}, \quad (4f)$$

$$z_0 = \hat{z}(t), \quad d = \hat{d}(t), \quad u_{-1} = u(t - T_s), \quad (4g)$$

where  $\hat{z}(t) \in \mathbb{R}^{n_z}$  is the lifted state estimate provided by the Kalman filter; and  $u_{-1} = u(t - T_s)$  is the input applied at the previous sampling instant, used to compute the first increment  $\Delta u_0$ . Following the receding-horizon principle, only the first element  $u_0^*$  of the optimal sequence is applied to the plant.

## 4. Methodology

Main drawback that we observe in a current Koopman MPC (offset-free, robust, or any other), is that the optimization is done without fully leveraging knowledge of the lifted space. Of course we are predicting in the lifted space, but in objective function we are using original space, that is created derived as linear transformation of the lifted space  $y_k = Cz_k$ . This transformation is not linear by nature  $y_k = h(z_k)$ , but to use linear MPC, we think we had no other choice. Some papers are not using this transformation directly, but after closer look, they are usually using equivalent transformations, for example expanding lifted space with measurements, but dynamic in lifted space is linear, so transformations to achieve back this measurements is linear. We will show that we can use the lifted space directly in the objective function, showing how to handle the lifted space and transformation of an objective function. We can fully eliminate the necessity of the back transformation used in objective function in MPC, but we still need them to compute the target for the lifted space. In this case and case of constraints, we use a first order Taylor expansion, to better approximate the transformation.

### 4.1. Linearization of Output Mapping

For the control problem, we are not able to completely eliminate the necessity of the transformation from the lifted space back to the original space. For this, we linearize the output mapping with first order Taylor expansion. For the purpose of this work we experimented with linearization at two different points: linearization at the previous target and linearization at the current state estimate.

$$y_k = h(z_k) \approx h(\bar{z}(t - T_s)) + J_{\bar{z}(t-T_s)}(z_k - \bar{z}(t - T_s)), \quad (5a)$$

$$y_k = h(z_k) \approx h(\hat{z}(t)) + J_{\hat{z}(t)}(z_k - \hat{z}(t)), \quad (5b)$$

where  $h : \mathbb{R}^{n_z} \rightarrow \mathbb{R}^{n_y}$  is the nonlinear output mapping;  $\bar{z}(t - T_s)$  is the steady-state lifted state computed by the target optimizer at the previous sampling instant;  $J_{\bar{z}(t-T_s)}$  and  $J_{\hat{z}(t)}$  are the Jacobians of  $h$  evaluated at  $\bar{z}(t - T_s)$  and  $\hat{z}(t)$ , respectively.

### 4.2. Target Optimization

The objective function of target estimation is the same as in the standard offset-free MPC. The main difference is in

The proposed target optimization problem is formulated as:

$$\min_{\bar{u}, \bar{z}, \bar{y}} (\bar{y} - y_{\text{ref}})^\top Q_y (\bar{y} - y_{\text{ref}}), \quad (6a)$$

$$\text{s.t.} \quad \bar{z} = A\bar{z} + B\bar{u}, \quad (6b)$$

$$\bar{y} = H(z_{\text{LP}})\bar{z} + h(z_{\text{LP}}) - H(z_{\text{LP}})z_{\text{LP}} + d, \quad (6c)$$

$$u_{\min} \leq \bar{u} \leq u_{\max}, \quad (6d)$$

$$y_{\min} \leq \bar{y} \leq y_{\max}, \quad (6e)$$

$$d = \hat{d}(t), \quad y_{\text{ref}} = y_{\text{sp}}, \quad z_{\text{LP}} = \bar{z}_{-1}, \quad (6f)$$

where  $H(z_{\text{LP}})$  is the Jacobian of the transformation  $h(z_{\text{LP}})$ .  $z_{\text{LP}}$  is a linearization point, the point where we approximate our function with first order Taylor expansion. We propose, that this point should be as close as possible to the  $\bar{z}$ . Therefor we are using the previously calculated target  $\bar{z}_{-1}$ . This allow us to better approximate the transformation of the lifted space to the original space and more precise target optimization.

#### 4.3. MPC

The proposed MPC problem is formulated as:

$$\min_{u_0, \dots, u_{N-1}} \sum_{k=0}^{N-1} (z_k - \bar{z})^\top Q_z (z_k - \bar{z}) + \sum_{k=0}^{N-1} \Delta u_k^\top Q_u \Delta u_k, \quad (7a)$$

$$\text{s.t.} \quad z_{k+1} = Az_k + Bu_k, \quad k \in \mathbb{N}_0^{N-1}, \quad (7b)$$

$$y_k = H(z_0)z_k + h(z_0) - H(z_0)z_0 + d, \quad k \in \mathbb{N}_0^{N-1}, \quad (7c)$$

$$\Delta u_k = u_k - u_{k-1}, \quad k \in \mathbb{N}_0^{N-1}, \quad (7d)$$

$$u_{\min} \leq u_k \leq u_{\max}, \quad k \in \mathbb{N}_0^{N-1}, \quad (7e)$$

$$y_{\min} \leq y_k \leq y_{\max}, \quad k \in \mathbb{N}_0^{N-1}, \quad (7f)$$

$$z_0 = \hat{z}(t), \quad d = \hat{d}(t), \quad u_{-1} = u(t - T_s) \quad (7g)$$

where  $H(z_0)$  is the Jacobian of the transformation  $h(z_0)$  at the point  $z_0$ . This linearization point is chosen to best approximate the surroundings of current operational point. The first order Taylor expansion is done to receive  $y_k$ , which is used only in the constrains of the system. In objective funtion, we use a tuning matrix  $Q_z$  to penalize the difference from lifted space target. We propose to derive this tuning matrix as follows:

$$Q_z = H(\bar{z})^\top Q_y H(\bar{z}). \quad (8)$$

This allows us to use and tune our problem using matrix  $Q_y$ , which is convenient and easily tuned. This is a significant change from the standard Koopman MPC, where the objective function is defined in the original space.

#### 4.4. Closed-Loop Control Structure

#### 4.5. State Estimation - Time Varying Kalman Filter

### 5. Block Diagonal Deep Koopman Transformation

**TODO: split this section into theory and simulation evaluation, becare notation conflict  $\bar{z}$**  During our work, we encounter a several problems with the suboptimal or infeasible solutions using Koopman MPC using Gurobi solver. For the enhancement of the performance, we are proposing a block diagonal structure of the matrix  $A$ . This transformation can be done after obtaining the Koopman operator, with finding the transformation matrix  $T$  that transforms the Koopman operator to the block diagonal form:

$$\bar{A} = T^{-1}AT. \quad (9)$$

This transformation is done by finding the eigenvalues and eigenvectors of the Koopman operator, and then grouping them into blocks. The transformation matrix  $T$  is constructed from the eigenvectors, and the block diagonal form  $\bar{A}$  is obtained by applying the transformation to the original Koopman operator  $A$ . Using this approach, we had to modify the rest of the model accordingly, using the transformation matrix  $T$  to transform the input and output matrices  $B$  and  $C$  as well as the lifted states  $z$ :

$$\bar{B} = T^{-1}B, \quad \bar{C} = CT, \quad \bar{z}_k = T^{-1}z_k. \quad (10)$$

When computing the Jackobian, we are transforming the original Jackobian  $H(z_k)$  to the block diagonal form  $\bar{H}(z_k)$  using the transformation matrix  $T$ :

$$\bar{H}(z_k) = H(z_k)T = H(T\bar{z}_k)T. \quad (11)$$

After this transformation we are able to enhance the performance in optimization as can be seen in Table 1. This table is done using the formulation of Koopman MPC which can be seen in Sec. 3. The table shows the decrease in a computational times. Also with this formulation we are able to obtain optimal solution 100% time during simulation. Contrary, the full Koopman MPC formulation is able to obtain optimal solution only 26.7% of the time



with the rest of the time giving warning, that the solution is suboptimal. We can see also increase in a control performance, which is measured by the objective function value, which is in optimal case around 6% lower. Also the settling time for both levels is lower.

Table 1: Comparison of structure of problem to solver time and control performance.

| Structure      | Time TE/MPC     | Optimal? [%] | OBJ   | ST $h_1$ | ST $h_2$ |
|----------------|-----------------|--------------|-------|----------|----------|
| Full           | 0.2392 / 2.3884 | 26.7         | 100.0 | 43       | 48       |
| Block Diagonal | 0.2191 / 1.6204 | 100.0        | 94.1  | 25       | 31       |

The second option how to obtain the block diagonal structure is to learn the Koopman operator directly in the block diagonal form. This can be done by limiting the number of learnable parameters in a Koopman matrix during training to those that are in the block diagonal form. This also speeds the training as less parameters are learned.

To obtain an optimal performance, which is necessary for the real time control, we are proposing to use the block diagonal structure of the Koopman operator. There is no necessity to use the full Koopman operator, as the block diagonal structure is able to capture the same dynamics of the system and favors the optimization performance. The way this structure is obtained, during or after the training, is not important, as long as the Koopman operator is in the block diagonal form.

## 6. Simulation Setup

## 7. Simulation Results

In the following, we make precise the output mappings sketched in the notes and list the combinations (setups) evaluated in the simulations.

### 7.1. Output mappings for target and prediction

We consider three alternatives for computing the steady-state output corresponding to a candidate steady-state lifted state  $\bar{z}_k$  and disturbance estimate  $\hat{d}_k$ :

$$\text{T1: } \bar{y}_k = C \bar{z}_k + \hat{d}_k, \quad (12a)$$

$$\text{T2: } \bar{y}_k = h(\bar{z}_{k-1}) + J_{\bar{z}_{k-1}} (\bar{z}_k - \bar{z}_{k-1}) + \hat{d}_k, \quad (12b)$$

$$\text{T3: } \bar{y}_k = h(\hat{z}_k) + J_{\hat{z}_k} (\bar{z}_k - \hat{z}_k) + \hat{d}_k. \quad (12c)$$

Here,  $h(\cdot)$  denotes the measurement/lifted-to-output map, and  $J_z$  its Jacobian evaluated at  $z$ .

For dynamic prediction over the horizon, with disturbance held at the current estimate  $\hat{d}_k$ , we use four alternatives for the output of a predicted lifted state  $z_{k+i}$ :

$$\text{D1: } y_{k+i} = C z_{k+i} + \hat{d}_k, \quad (13a)$$

$$\text{D2: } y_{k+i} = h(\bar{z}_{k-1}) + J_{\bar{z}_{k-1}} (\bar{z}_{k+i} - \bar{z}_{k-1}) + \hat{d}_k, \quad (13b)$$

$$\text{D3: } y_{k+i} = h(\hat{z}_k) + J_{\hat{z}_k} (\bar{z}_{k+i} - \hat{z}_k) + \hat{d}_k, \quad (13c)$$

$$\text{D4: } y_{k+i} = h(\bar{z}_k) + J_{\bar{z}_k} (z_{k+i} - \bar{z}_k) + \hat{d}_k. \quad (13d)$$

## 7.2. Simulation Results for Two Tanks System

We index target mappings by T1–T3 and dynamic mappings by D1–D4 as in (12)–(13). The simulation setups below pair one target mapping with one dynamic mapping:

Table 2: Simulation setups matrix, with  $Q_y$  much higher  $Q_u$ ;  $Q_y = \text{diag}(5)$   $Q_u = \text{diag}(0.1)$ . Columns: dynamic mappings (D1–D4). Rows: target mappings (T1–T3). Fill cells with the identifier/value for each tested combination. NMPC benchmark is 100.

| TM | DM     |        |        |        |
|----|--------|--------|--------|--------|
|    | D1     | D2     | D3     | D4     |
| T1 | 124.19 | x      | x      | x      |
| T2 | x      | 104.86 | 107.47 | 103.79 |
| T3 | x      | 105.34 | 108.11 | 103.89 |

Table 3: Simulation setups matrix, with  $Q_y$  and  $Q_u$  with equal diagonal values (5). Columns: dynamic mappings (D1–D4). Rows: target mappings (T1–T3). NMPC benchmark is 100.

| TM | DM     |        |        |        |
|----|--------|--------|--------|--------|
|    | D1     | D2     | D3     | D4     |
| T1 | 112.90 | x      | x      | x      |
| T2 | x      | 108.86 | 106.44 | 107.52 |
| T3 | x      | 106.24 | 108.94 | 105.33 |

Table 4: Simulation setups matrix, with  $Q_y$  much lower  $Q_u$ ;  $Q_y = \text{diag}(0.1)$   $Q_u = \text{diag}(5)$ . Columns: dynamic mappings (D1–D4). Rows: target mappings (T1–T3). NMPC benchmark is 100.

| TM | DM     |        |        |        |
|----|--------|--------|--------|--------|
|    | D1     | D2     | D3     | D4     |
| T1 | 147.49 | x      | x      | x      |
| T2 | x      | 189.52 | 176.33 | 190.96 |
| T3 | x      | 143.29 | 143.86 | 143.47 |

### 7.3. Simulation Results for Two Reactor System

We index target mappings by T1–T3 and dynamic mappings by D1–D4 as in (12)–(13). Table 5 compares previous diagonal entries (CT, T2D2, T3D3) with reruns from the current scripts.

Table 5: Two-reactor CSTR: normalized closed-loop objective with NMPC benchmark 100. Diagonal combinations CT ( $T1 \times D1$ ), T2D2 ( $T2 \times D2$ ), and T3D3 ( $T3 \times D3$ ). *Previous*: entries from the earlier full  $3 \times 4$  TM/DM matrices. *Updated*:  $100 J/J_{\text{NMPC}}$  from `control_cstr` (CT.py, T2D2.py, T3D3.py) with the same weighting as tuning I; em dashes mark tunings II–III not yet rerun.

| Tuning                                                 | Previous |       |       | Updated |       |       |
|--------------------------------------------------------|----------|-------|-------|---------|-------|-------|
|                                                        | CT       | T2D2  | T3D3  | CT      | T2D2  | T3D3  |
| I: $Q_y = \text{diag}(5)$ , $Q_u = \text{diag}(0.1)$   | 365.0    | 251.5 | 265.1 | 314.9   | 260.5 | 285.7 |
| II: $Q_y = Q_u = I$                                    | 184.5    | 146.2 | 153.9 | —       | —     | —     |
| III: $Q_y = \text{diag}(0.1)$ , $Q_u = \text{diag}(5)$ | 130.4    | 112.1 | 116.1 | —       | —     | —     |

## 8. Pasteurization Unit: Experimental Device

The experimental validation of the proposed control strategy is conducted on a laboratory-scale pasteurization unit, a thermal processing system designed for educational and research purposes. This apparatus, depicted in Figure 2, represents a complex multivariable thermal system commonly encountered in food processing industries, where precise temperature regulation is crucial for product quality and safety.



Figure 2: Pasteurization unit experimental apparatus.

### 8.1. System Configuration

The pasteurization unit comprises a tubular heat exchanger configuration with counter-current flow arrangement, as illustrated schematically in Figure 3. The system consists of two primary fluid circuits: the product circuit and the heating medium circuit. The product fluid is pumped through a helical tube nested within a cylindrical shell, where it exchanges thermal energy with hot water circulating in the shell side.

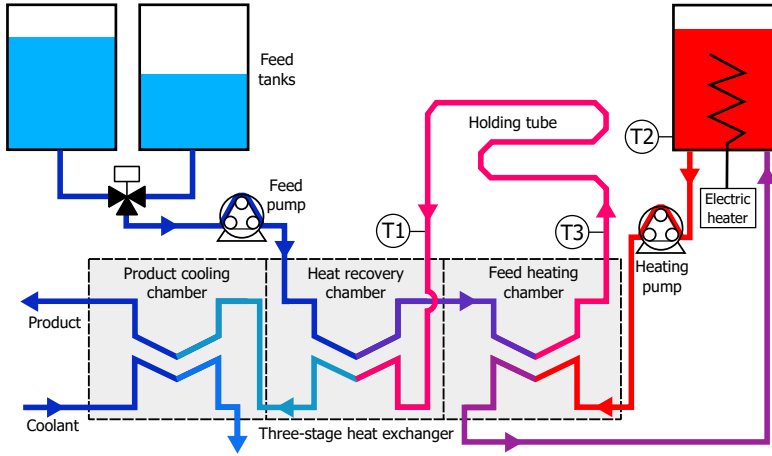


Figure 3: Schematic representation of the heat exchanger configuration.

The unit features eight manipulated variables that can be adjusted for process control:

- **Pump1:** Product feed flow rate control (%)
- **Pump2:** Heating medium circulation rate (%)
- **Heater:** Electric heater power for the heating medium (%)
- **DV:** Diversion valve position
- **FSV:** Feed selection valve
- **CWV:** Cooling water valve position
- **TAFV, TBFV:** Tank flow control valves

The system is instrumented with multiple sensors providing real-time measurements, including four temperature sensors (**T1–T4**) positioned at strategic locations along the product and heating circuits, a flow meter (**F1**) measuring product flow rate, a level sensor (**L1**) for tank monitoring, and additional auxiliary measurements for comprehensive process monitoring.

### 8.2. Control Configuration

For the experimental implementation, the control objective focuses on regulating the outlet product temperature (T4) to track desired setpoint trajectories. The primary manipulated variable is the heating medium flow rate (Pump2), which directly influences the thermal energy transferred to the product stream. Additionally, a cascade control structure is employed for the heater, where the electric heater power is modulated to maintain a constant heating medium inlet temperature (T2), providing a stable thermal source. The product feed rate (Pump1) acts as a measurable disturbance, creating varying thermal loads that challenge the controller’s disturbance rejection capabilities.

The system exhibits typical characteristics of thermal processes, including significant time delays, nonlinear behavior due to temperature-dependent fluid properties, and strong coupling between flow rates and temperatures. These features make it an ideal testbed for evaluating advanced control strategies such as the proposed offset-free Koopman MPC framework.

### 8.3. Data Acquisition and Control Platform

The experimental apparatus is integrated with a distributed control system accessible through the ELab platform, which provides remote monitoring and control capabilities via a web-based SCADA interface. The control algorithms are implemented in MATLAB/Simulink environment with a sampling period of 1 s, enabling real-time control execution. All process variables are logged continuously for post-experiment analysis and performance evaluation.

## 9. Experimental Results

## 10. Discussion

## 11. Conclusion

## Appendix A. System Models

In this appendix, we provide detailed mathematical descriptions of the two benchmark nonlinear systems used to evaluate the proposed offset-free Koopman MPC schemes: a two-tank system with hydraulic flows and a two-reactor CSTR series with recycle and parallel reactions.

### Appendix A.1. Two Tanks System

The two-tank system consists of two interconnected tanks with liquid heights  $h_1$  and  $h_2$ . Each tank receives an inflow (control inputs  $u_1$  and  $u_2$ , respectively), and liquid can flow between tanks and out of the system through gravity-driven mechanisms. The continuous-time nonlinear dynamics are described by:

$$\frac{dh_1}{dt} = \frac{1}{A_1} (u_1 - q_{12}), \quad (\text{A.1a})$$

$$\frac{dh_2}{dt} = \frac{1}{A_2} (u_2 + q_{12} - q_{\text{out}}), \quad (\text{A.1b})$$

where  $A_1$  and  $A_2$  are the cross-sectional areas of tanks 1 and 2, respectively. The inter-tank flow  $q_{12}$  and the outlet flow  $q_{\text{out}}$  follow nonlinear hydraulic relationships:

$$q_{12} = k_1 \operatorname{sgn}(h_1 - h_2) \sqrt{|h_1 - h_2|}, \quad (\text{A.2})$$

$$q_{\text{out}} = k_2 \sqrt{h_2}, \quad (\text{A.3})$$

with  $k_1$  and  $k_2$  being flow coefficients. The outputs are the measured tank heights:

$$y = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}. \quad (\text{A.4})$$

The system is discretized with a sampling time  $T_s = 1$  s using a Runge–Kutta integration scheme. The physical parameters and operational constraints used in the simulations are summarized in Table A.6.

### Appendix A.2. Two Reactor System

The two-reactor system consists of two continuous stirred tank reactors (CSTRs) in series with a recycle loop from the second reactor back to the first. Each reactor is equipped with jacket cooling. The system models a classic selectivity problem with parallel competing reactions on reactant A.

#### Appendix A.2.1. Chemical Reactions

Three reaction pathways are considered:

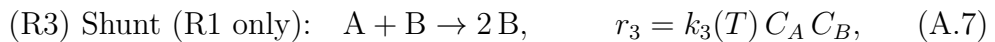


Table A.6: Two-tank system parameters and operational constraints.

| Parameter      | Description                    | Value                                 |
|----------------|--------------------------------|---------------------------------------|
| $A_1$          | Cross-sectional area of tank 1 | $1.0 \text{ m}^2$                     |
| $A_2$          | Cross-sectional area of tank 2 | $1.0 \text{ m}^2$                     |
| $k_1$          | Inter-tank flow coefficient    | $0.5 \text{ m}^{2.5}/\text{s}$        |
| $k_2$          | Outlet flow coefficient        | $0.3 \text{ m}^{2.5}/\text{s}$        |
| $h_1^0, h_2^0$ | Initial heights                | $0.5 \text{ m}$                       |
| $u_1^0, u_2^0$ | Initial inflows                | $0.0705, 0.4759 \text{ m}^3/\text{s}$ |
| $u_1 \in$      | Inflow constraint (tank 1)     | $[0.0, 0.5] \text{ m}^3/\text{s}$     |
| $u_2 \in$      | Inflow constraint (tank 2)     | $[0.0, 1.0] \text{ m}^3/\text{s}$     |
| $T_s$          | Sampling time                  | $1.0 \text{ s}$                       |

where the temperature-dependent rate constants follow Arrhenius kinetics:

$$k_i(T) = k_{i,0} \exp\left(-\frac{E_i}{RT}\right), \quad i = 1, 2, 3. \quad (\text{A.8})$$

The autocatalytic shunt reaction (R3) occurs only in the first reactor.

#### *Appendix A.2.2. Dynamic Model*

The system has eight states:

$$x = [C_{A,1} \quad T_1 \quad C_{A,2} \quad T_2 \quad C_{B,1} \quad C_{B,2} \quad C_{U,1} \quad C_{U,2}]^\top, \quad (\text{A.9})$$

where  $C_{A,i}$ ,  $C_{B,i}$ , and  $C_{U,i}$  denote concentrations of species A, B, and U in reactor  $i$ , and  $T_i$  is the temperature. The four control inputs are:

$$u = [F \quad L \quad T_{c,1} \quad T_{c,2}]^\top, \quad (\text{A.10})$$

with  $F$  the fresh feed flow rate,  $L$  the recycle flow rate, and  $T_{c,i}$  the jacket coolant temperatures.



The continuous-time dynamics are:

$$\frac{dC_{A,1}}{dt} = \frac{1}{V_1} [C_{A,0} F + L C_{A,2} - (F + L) C_{A,1}] - 2r_{1,1} - r_{2,1} - r_{3,1}, \quad (\text{A.11a})$$

$$\frac{dC_{B,1}}{dt} = \frac{1}{V_1} [C_{B,0} F + L C_{B,2} - (F + L) C_{B,1}] + r_{1,1} + r_{3,1}, \quad (\text{A.11b})$$

$$\frac{dC_{U,1}}{dt} = \frac{1}{V_1} [C_{U,0} F + L C_{U,2} - (F + L) C_{U,1}] + r_{2,1}, \quad (\text{A.11c})$$

$$\begin{aligned} \frac{dT_1}{dt} = & \frac{1}{V_1} [T_0 F + L T_2 - (F + L) T_1] - \frac{U_1 A_1}{V_1 \rho c_p} (T_1 - T_{c,1}) \\ & + \frac{1}{\rho c_p} [(-\Delta H_1) r_{1,1} + (-\Delta H_2) r_{2,1} + (-\Delta H_3) r_{3,1}], \end{aligned} \quad (\text{A.11d})$$

$$\frac{dC_{A,2}}{dt} = \frac{F + L}{V_2} [C_{A,1} - C_{A,2}] - 2r_{1,2} - r_{2,2}, \quad (\text{A.11e})$$

$$\frac{dC_{B,2}}{dt} = \frac{F + L}{V_2} [C_{B,1} - C_{B,2}] + r_{1,2}, \quad (\text{A.11f})$$

$$\frac{dC_{U,2}}{dt} = \frac{F + L}{V_2} [C_{U,1} - C_{U,2}] + r_{2,2}, \quad (\text{A.11g})$$

$$\begin{aligned} \frac{dT_2}{dt} = & \frac{F + L}{V_2} [T_1 - T_2] - \frac{U_2 A_2}{V_2 \rho c_p} (T_2 - T_{c,2}) \\ & + \frac{1}{\rho c_p} [(-\Delta H_1) r_{1,2} + (-\Delta H_2) r_{2,2}], \end{aligned} \quad (\text{A.11h})$$

where  $r_{j,i}$  denotes the rate of reaction  $j$  in reactor  $i$ . The outputs are all eight states:

$$y = x. \quad (\text{A.12})$$

The system is discretized with sampling time  $T_s = 1$  s using a Runge–Kutta integration scheme. Physical parameters, kinetic constants, and operational constraints are listed in Tables A.7–A.9.

Table A.7: CSTR series physical and thermodynamic parameters.

| Parameter          | Description                                  | Value                                                        |
|--------------------|----------------------------------------------|--------------------------------------------------------------|
| $V_1$              | Volume of reactor 1                          | $1.0 \times 10^{-3} \text{ m}^3$                             |
| $V_2$              | Volume of reactor 2                          | $2.0 \times 10^{-3} \text{ m}^3$                             |
| $U_1 A_1$          | Heat transfer coefficient $\times$ area (R1) | $0.461 \text{ kJ}/(\text{s}\cdot\text{K})$                   |
| $U_2 A_2$          | Heat transfer coefficient $\times$ area (R2) | $0.732 \text{ kJ}/(\text{s}\cdot\text{K})$                   |
| $\rho$             | Liquid density                               | $1050 \text{ kg}/\text{m}^3$                                 |
| $c_p$              | Specific heat capacity                       | $3.766 \text{ kJ}/(\text{kg}\cdot\text{K})$                  |
| $R$                | Universal gas constant                       | $8.3145 \times 10^{-3} \text{ kJ}/(\text{mol}\cdot\text{K})$ |
| $C_{A,0}$          | Feed concentration of A                      | $97.35 \text{ mol}/\text{m}^3$                               |
| $C_{B,0}, C_{U,0}$ | Feed concentrations of B, U                  | $0.0 \text{ mol}/\text{m}^3$                                 |
| $T_0$              | Feed temperature                             | $298.0 \text{ K}$                                            |

Table A.8: Arrhenius kinetic parameters for reactions R1–R3.

| Reaction                                          | Parameter                  | Pre-exponential                                         | Activation Energy            | Heat of Reaction             |
|---------------------------------------------------|----------------------------|---------------------------------------------------------|------------------------------|------------------------------|
| R1: $2 \text{ A} \rightarrow \text{B}$            | $k_{1,0}, E_1, \Delta H_1$ | $1.0 \times 10^5 \text{ m}^3/(\text{mol}\cdot\text{s})$ | $45.0 \text{ kJ}/\text{mol}$ | $60.0 \text{ kJ}/\text{mol}$ |
| R2: $\text{A} \rightarrow \text{U}$               | $k_{2,0}, E_2, \Delta H_2$ | $9.8 \times 10^9 \text{ s}^{-1}$                        | $70.0 \text{ kJ}/\text{mol}$ | $40.0 \text{ kJ}/\text{mol}$ |
| R3: $\text{A} + \text{B} \rightarrow 2 \text{ B}$ | $k_{3,0}, E_3, \Delta H_3$ | $5.0 \times 10^4 \text{ m}^3/(\text{mol}\cdot\text{s})$ | $55.0 \text{ kJ}/\text{mol}$ | $60.0 \text{ kJ}/\text{mol}$ |

Table A.9: CSTR series operational constraints and initial conditions.

| Parameter              | Description                  | Value/Range                                                                  |
|------------------------|------------------------------|------------------------------------------------------------------------------|
| $F \in$                | Fresh feed flow rate         | $[0.0, 2.0 \times 10^{-4}] \text{ m}^3/\text{s}$                             |
| $L \in$                | Recycle flow rate            | $[0.0, 2.0 \times 10^{-4}] \text{ m}^3/\text{s}$                             |
| $T_{c,1}, T_{c,2} \in$ | Coolant temperatures         | $[280.0, 330.0] \text{ K}$                                                   |
| $x_0$                  | Initial state (steady state) | Computed for $u_0 = [1.5 \times 10^{-4}, 7.5 \times 10^{-5}, 290, 290]^\top$ |
| $T_s$                  | Sampling time                | $1.0 \text{ s}$                                                              |